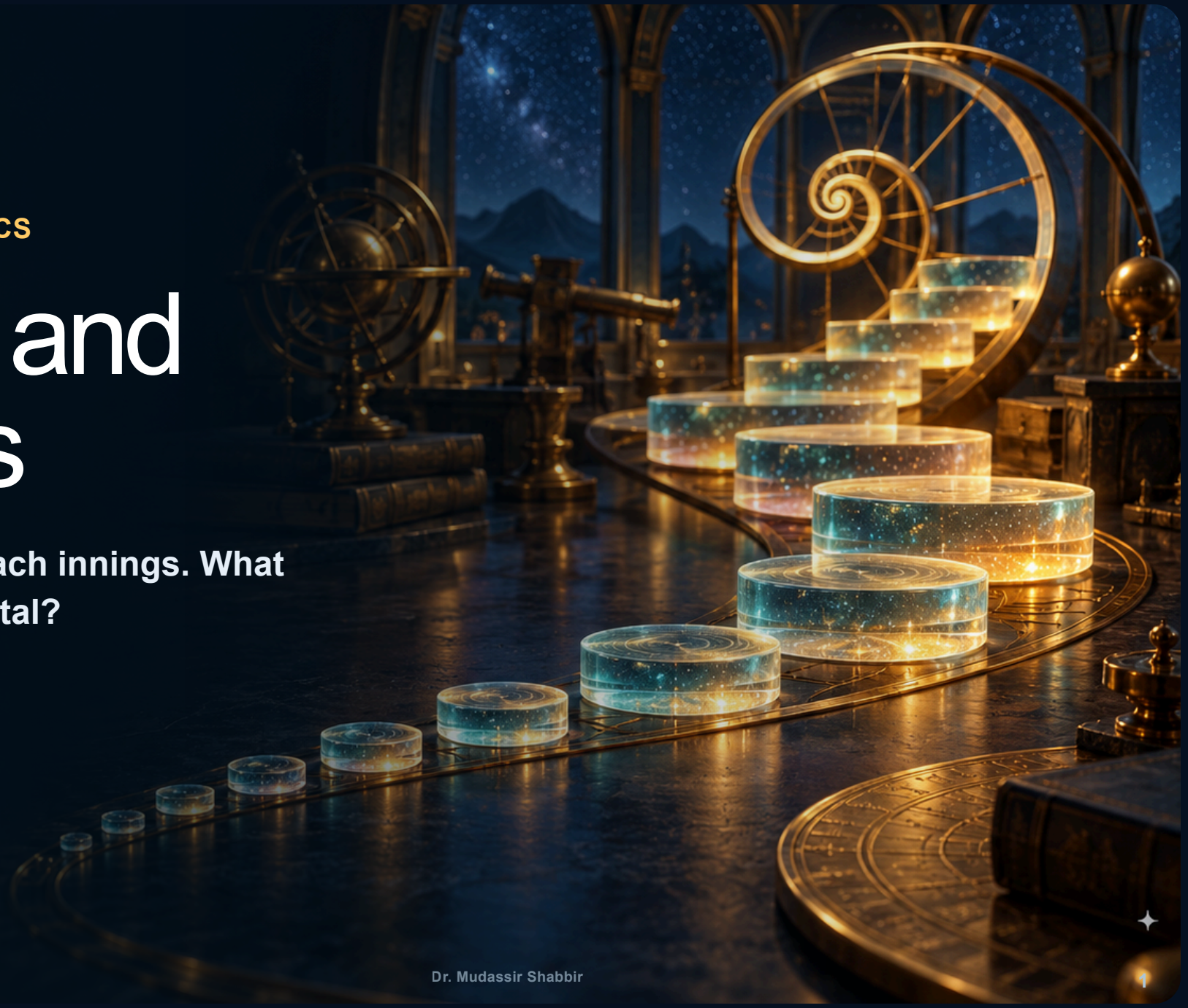
LECTURE 18 · DISCRETE MATHEMATICS

Sequences and summations

Scores start at 0 and rise by 5 each innings. What is the 25th score? What is the total?



A sequence maps an index to a term

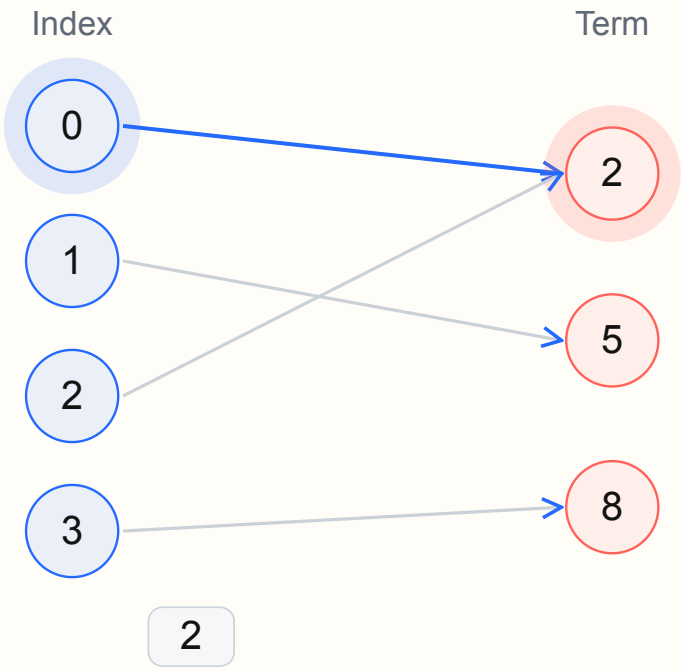
Each integer index n in the domain has one value a_n.

Index

$0, 1, 2, \ldots$ **or** $1, 2, 3, \ldots$

Term

Values may repeat.



$0 \mapsto 2$: the first term

A sequence keeps order and repetition; a set does not.

A sequence maps an index to a term

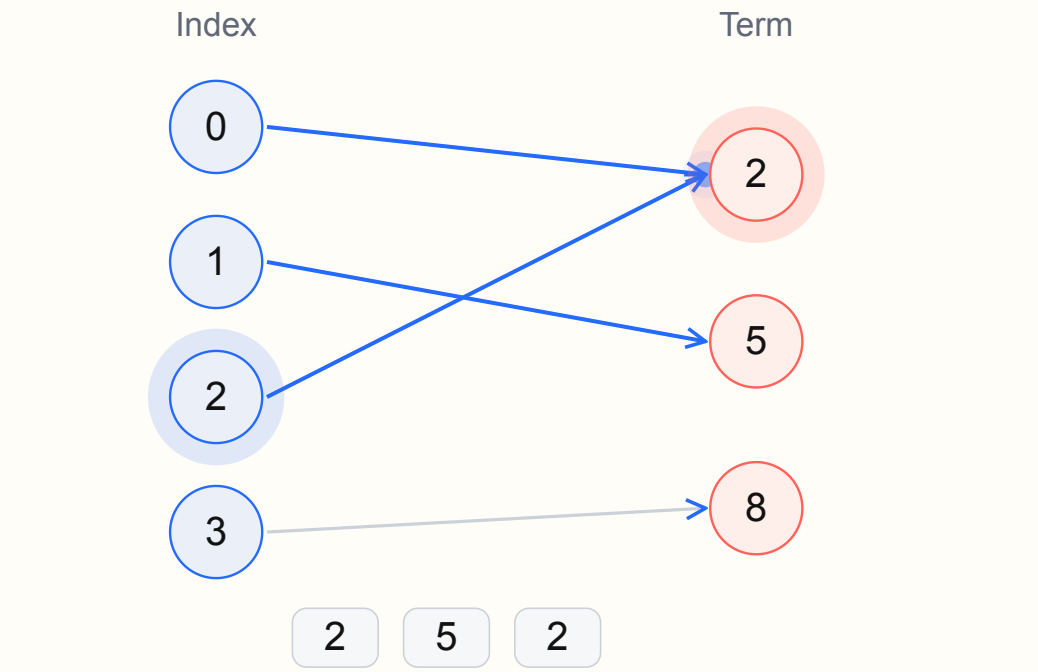
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Index

$0, 1, 2, \ldots$ **or** $1, 2, 3, \ldots$

Term

Values may repeat.



$2 \mapsto 2$: a value may repeat

A sequence keeps order and repetition; a set does not.

A sequence maps an index to a term

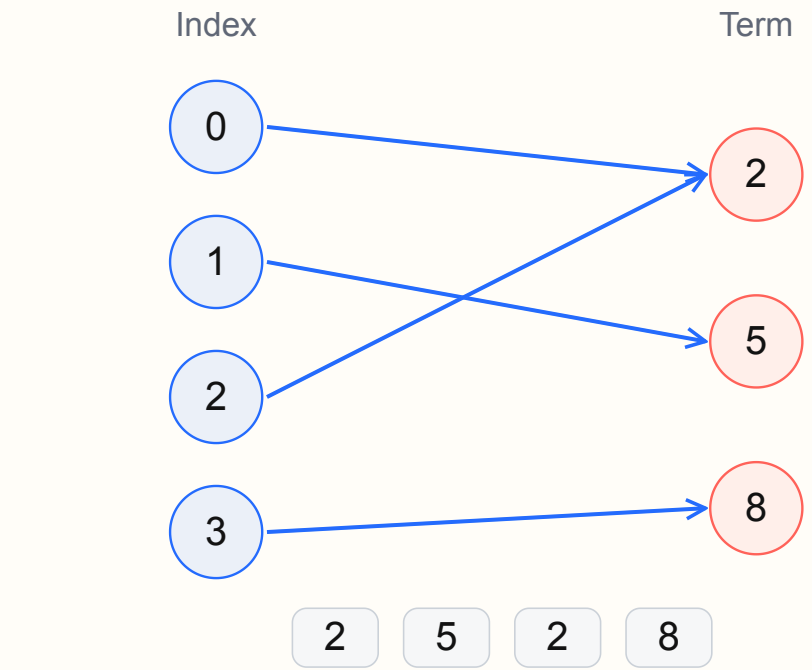
Each integer index n in the domain has one value a_n.

Index

$0, 1, 2, \ldots$ **or** $1, 2, 3, \ldots$

Term

Values may repeat.



$a_0, a_1, a_2, a_3 = 2, 5, 2, 8$

A sequence keeps order and repetition; a set does not.

EXERCISE 1 · PATTERNS NEED A RULE

The same prefix can fit different rules

Both rules begin $2, 4, 6$ at indices $1, 2, 3$.

01 **For $a_n = 2n$, find a_4.**

$a_4 = 8$.

02 **For $b_n = n^3 - 6n^2 + 13n - 6$, find b_4.**

$b_4 = 14$.

03 **What must a pattern question specify?**

A rule or an assumption about how the terms continue.

Strings are finite sequences

A string records an ordered list of symbols from an alphabet.

A bit string

$$0, 1, 1, 0, 1$$

Length 5; alphabet $\{0, 1\}$.

The empty string

$$\lambda$$

Length 0; no terms.

Order and repetition matter even when the terms are not numbers.

Geometric progressions

Each term is the previous term multiplied by a fixed ratio r.

$$a, ar, ar^2, \ldots, ar^n, \ldots$$

Initial term and rule

$$a_0 = a, \qquad a_n = ar^n \quad (n \geq 1)$$

Consecutive terms

$$a_{n+1} = ra_n$$

For $r = 0$, keep $a_0 = a$ and use the multiplication rule thereafter.

EXAMPLE · GEOMETRIC

Geometric progressions examples

All three sequences start at $n = 0$.

Rule	Initial term a	Ratio r	First terms
$(-1)^n$	1	-1	$1, -1, 1, -1, \ldots$
$2 \cdot 5^n$	2	5	$2, 10, 50, 250, \ldots$
$6(1/3)^n$	6	$1/3$	$6, 2, \frac{2}{3}, \frac{2}{9}, \ldots$

A fixed ratio can produce alternating, growing, or shrinking terms.

EXERCISE 2

Geometric rules: shrinking and alternating

Both sequences are geometric and start at $n = 0$.

01 **For $8, 4, 2, 1, \ldots$, find a_n and a_5.**

$a_n = 8(1/2)^n$ **and** $a_5 = 1/4$.

---

02 **For $2, -4, 8, -16, \ldots$, find b_n and b_4.**

$b_n = 2(-2)^n$ **and** $b_4 = 32$.

Arithmetic progressions

Each term is the previous term plus a fixed difference d.

$$a, a+d, a+2d, \ldots, a+nd, \ldots$$

Initial term and rule

$$a_0 = a, \qquad a_n = a + nd \quad (n \geq 0)$$

Consecutive terms

$$a_{n+1} - a_n = d$$

The difference d may be positive, zero, or negative.

EXAMPLE · ARITHMETIC

Arithmetic progressions examples

Both sequences start at $n = 0$.

Add 4: $a = -1$, $d = 4$

$$s_n = -1 + 4n$$

$-1, 3, 7, 11, \ldots$

Subtract 3: $a = 7$, $d = -3$

$$t_n = 7 - 3n$$

$7, 4, 1, -2, \ldots$

Check the initial term, then the difference between consecutive terms.

EXERCISE 3

Negative steps, different starting indices

Assume $12, 8, 4, 0, \ldots$ continues by subtracting 4.

01 **First term a_0: find d and a_n.**

$d = -4$, **so** $a_n = 12 - 4n$, $n \geq 0$.

02 **First term b_1: find b_n and b_6.**

$b_n = 12 - 4(n-1)$, $n \geq 1$; $b_6 = -8$.

03 **Does changing the first index change the direction?**

No. Both sequences are strictly decreasing.

Harmonic progressions

A sequence of nonzero terms is harmonic when its reciprocals form an arithmetic progression.

Arithmetic reciprocals

$$c, c+d, c+2d, \ldots$$

Harmonic terms

$$a_n = \frac{1}{c+nd}, \qquad n \geq 0$$

The denominator must be nonzero at every index in the domain.

EXAMPLE · HARMONIC

A reciprocal sequence: $a_n = 1/n$

Our first sequence example starts at $n = 1$.

Terms

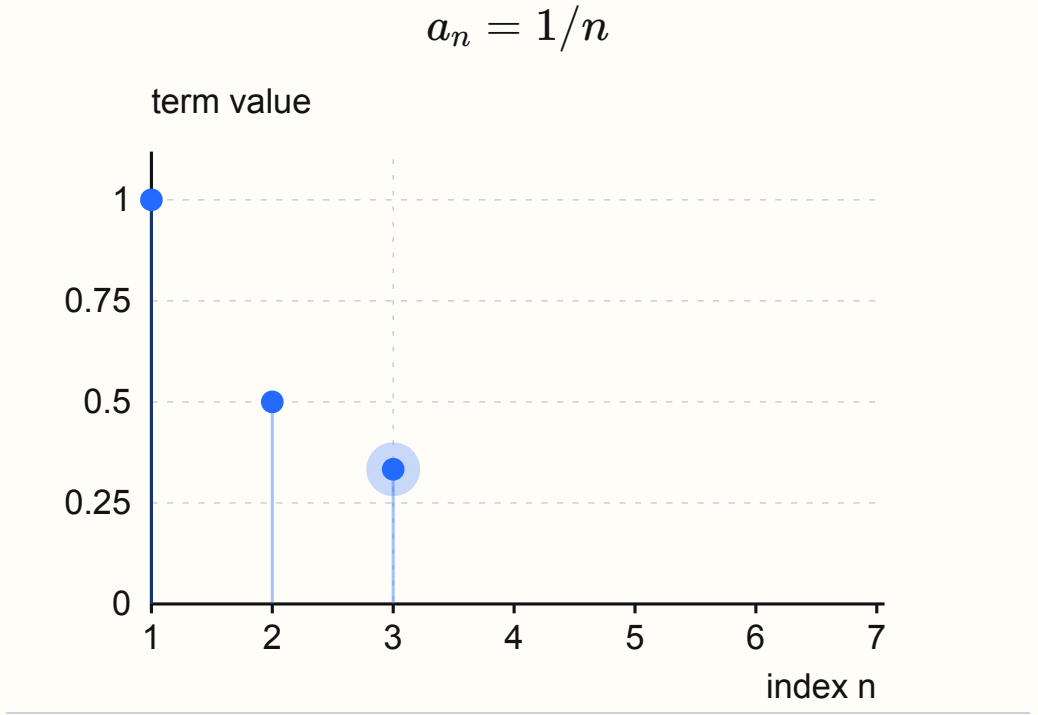
$$1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \dots$$

Reciprocals

$$1, 2, 3, 4, \dots$$

A later term

$$a_5 = \frac{1}{5}$$



$a_3 = 1/3 \qquad 1/a_3 = 3$

The reciprocals have constant difference 1.

EXAMPLE · HARMONIC

A reciprocal sequence: $a_n = 1/n$

Our first sequence example starts at $n = 1$.

Terms

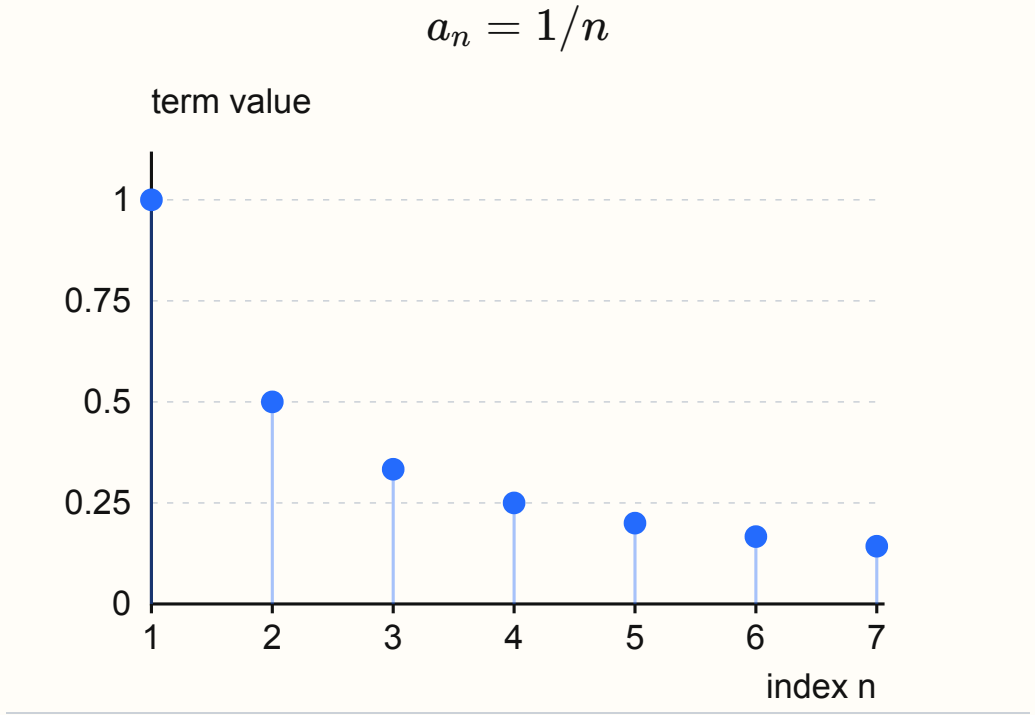
$$1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \ldots$$

Reciprocals

$$1, 2, 3, 4, \ldots$$

A later term

$$a_5 = \frac{1}{5}$$



$a_n = 1/n$: smaller terms, larger reciprocals.

The reciprocals have constant difference 1.

EXERCISE 4

Arithmetic, geometric, or harmonic?

Assume each displayed pattern continues. Identify the rule.

01 $4, 7, 10, 13, \ldots$

Arithmetic: common difference 3.

---

02 $5, 15, 45, 135, \ldots$

Geometric: common ratio 3.

---

03 $1/3, 1/6, 1/12, 1/24, \ldots$

Geometric: common ratio $1/2$. The reciprocals $3, 6, 12, 24$ have no common difference.

Adding 2 versus multiplying by 2

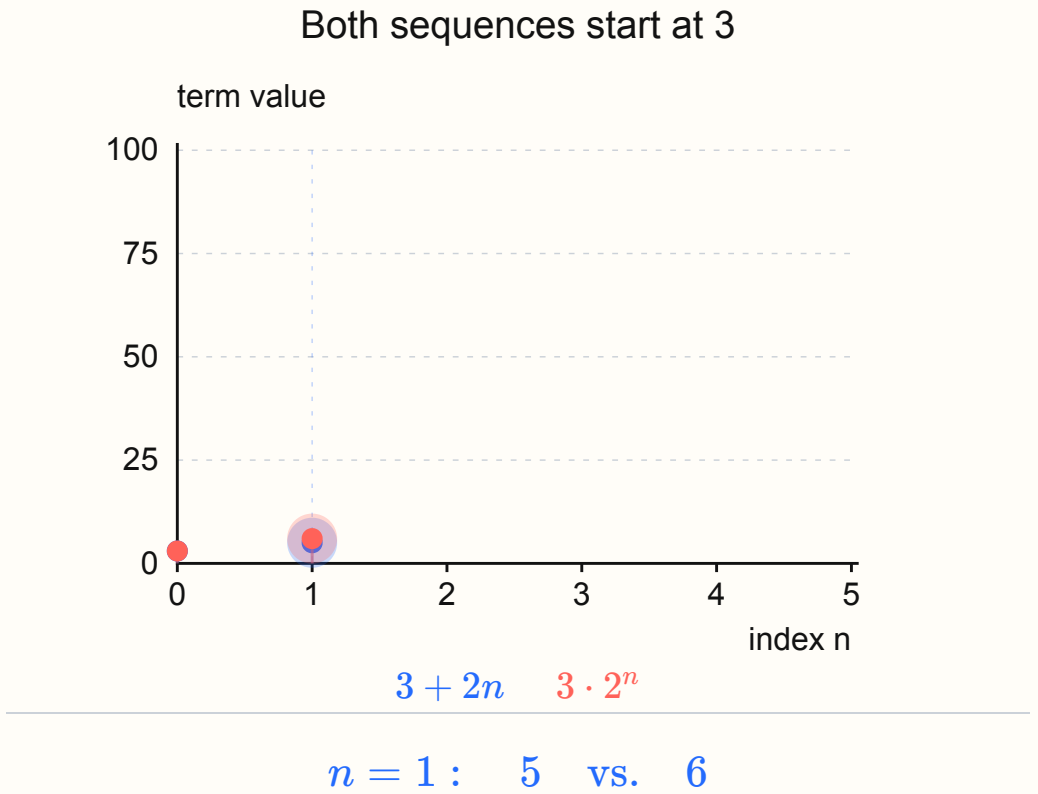
Both sequences start at 3, with $n = 0$.

Arithmetic: add 2

$$a_n = 3 + 2n$$

Geometric: double

$$b_n = 3 \cdot 2^n$$



$n = 1 : \quad 5 \quad \text{vs.} \quad 6$

At index 5: $a_5 = 13$, **while** $b_5 = 96$.

Adding 2 versus multiplying by 2

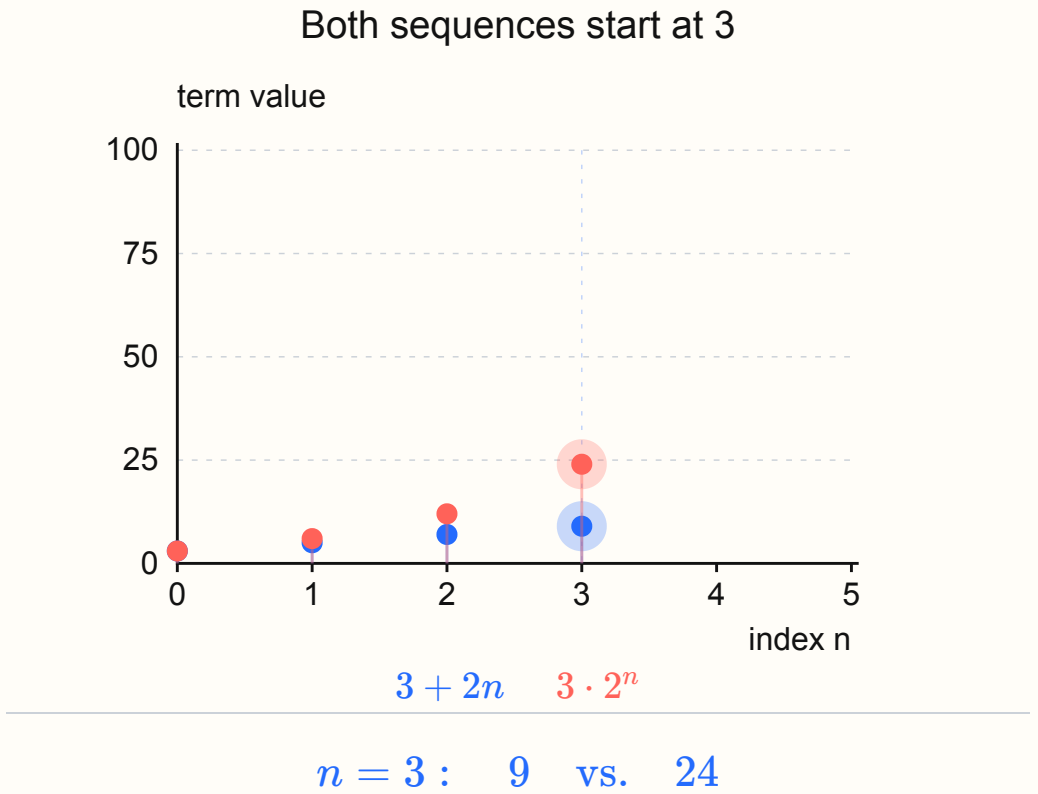
Both sequences start at 3, with $n = 0$.

Arithmetic: add 2

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$n = 3$: 9 vs. 24

At index 5: $a_5 = 13$, **while** $b_5 = 96$.

Adding 2 versus multiplying by 2

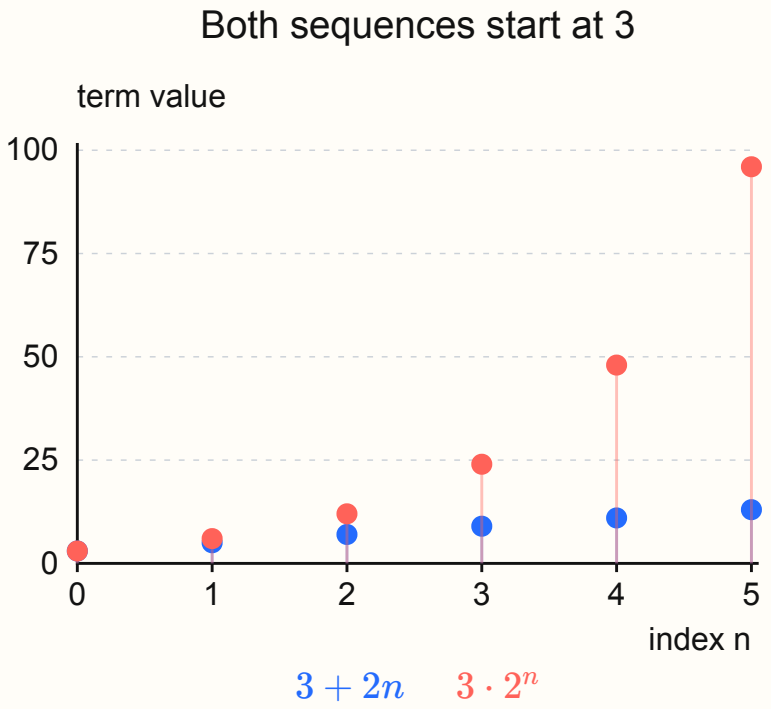
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$$a_n = 3 + 2n$$

Geometric: double

$$b_n = 3 \cdot 2^n$$



At $n = 5$: 13 versus 96.

At index 5: $a_5 = 13$, while $b_5 = 96$.

Monotonicity: can equal terms occur?

The inequality must hold at every consecutive pair.

Property	Condition	Example
Strictly increasing	$a_{n+1} > a_n$	1, 2, 3, 4, …
Nondecreasing	$a_{n+1} \geq a_n$	1, 1, 2, 2, …
Strictly decreasing	$a_{n+1} < a_n$	8, 4, 2, 1, …
Nonincreasing	$a_{n+1} \leq a_n$	4, 4, 3, 3, …

“Strictly” excludes equality. Rosen calls nondecreasing “increasing”.

EXERCISE 5

Cricket: the score in innings 25

First innings: 0 runs.
Each later innings: 5 more runs.

01 **Find the score a_n.**

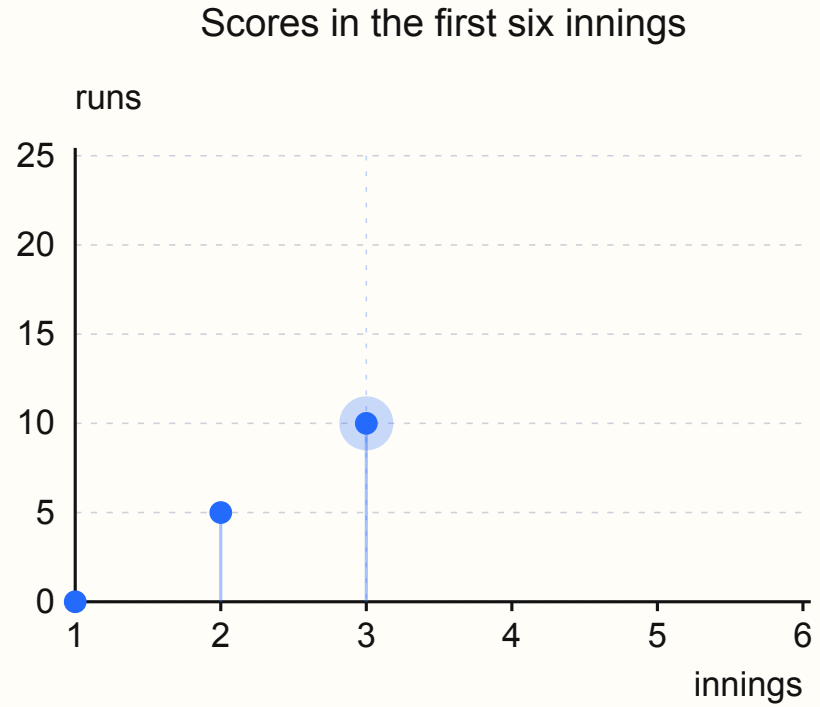
$a_n = 5(n-1), n \geq 1.$

02 **Find a_{25}.**

$5 \cdot 24 = 120$ **runs.**

03 **Predict the mean after 25 innings.**

60 runs; we will justify it using a sum.



$a_3 = 0 + (3-1) \cdot 5 = 10$

EXERCISE 5

Cricket: the score in innings 25

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01 **Find the score a_n.**

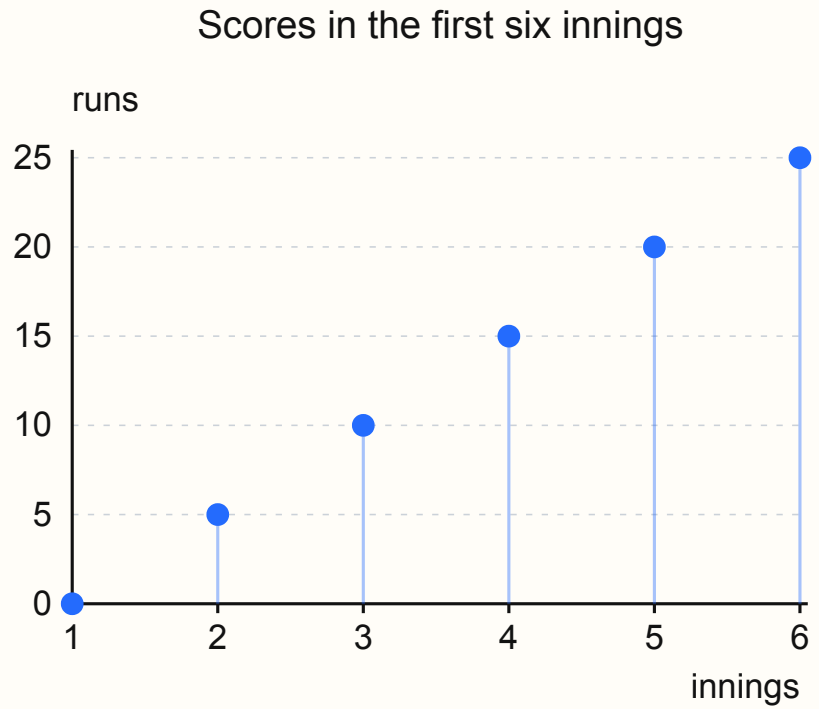
$a_n = 5(n-1), n \geq 1.$

02 **Find a_{25}.**

$5 \cdot 24 = 120$ **runs.**

03 **Predict the mean after 25 innings.**

60 runs; we will justify it using a sum.



$a_n = 5(n-1)$: five more runs each innings.

A sum adds terms across an index range

A finite series adds consecutive terms of a sequence.

$$\sum_{k=m}^{n} a_k = a_m + a_{m+1} + \cdots + a_n$$

Number of terms

$n - m + 1$, **for** $m \le n$

One term versus a total

a_4 **versus** $\sum_{k=1}^{4} a_k$

The index k is a placeholder; another letter gives the same sum.

EXERCISE 6

Expand the sum and count its terms

Write out each sum before evaluating it.

01 $\displaystyle\sum_{k=0}^{3} 2^k$

$1 + 2 + 4 + 8 = 15.$

02 $\displaystyle\sum_{k=1}^{3} 2^k$

$2 + 4 + 8 = 14.$

03 $\displaystyle\sum_{k=2}^{5} k$

$2 + 3 + 4 + 5 = 14.$

Reindex the terms and the limits together

Let $k = j + 3$. The values being added stay the same.

Original indices

$$\sum_{j=1}^{6} (j+3)^3$$

New indices

$$\sum_{k=4}^{9} k^3$$

Both sums are $4^3 + 5^3 + \cdots + 9^3$: six terms.

Linearity lets us split a sum

The constants c and d do not depend on k.

$$\sum_{k=m}^{n}(ca_k + db_k) = c\sum_{k=m}^{n} a_k + d\sum_{k=m}^{n} b_k$$

Separate a linear expression

$$\sum_{k=1}^{n}(2k+3) = 2\sum_{k=1}^{n} k + 3n$$

Count the constant once per index

$$\sum_{k=m}^{n} c = (n-m+1)c$$

A sum of products is generally not the product of two sums.

EXERCISE 7

Rewrite before you calculate

Keep the same terms; choose notation that makes the work easier.

01 **Rewrite $\sum_{j=2}^{5}(j-1)^2$ with $k=j-1$.**

$$\sum_{k=1}^{4} k^2 = 1+4+9+16 = 30.$$

---

02 **Simplify $\sum_{k=1}^{n}(4k-2)$.**

$$4\sum_{k=1}^{n} k - 2n.$$

A double sum adds one row at a time

Hold i fixed while the inner index j runs through its values.

$$\sum_{i=1}^{4}\sum_{j=1}^{3} ij$$

Inner sum

$$i + 2i + 3i = 6i$$

Outer sum

$$6 + 12 + 18 + 24 = 60$$

	j = 1	j = 2	j = 3	Inner sums
i = 1	1	2	3	
i = 2	2	4	6	
i = 3	3	6	9	
i = 4	4	8	12	

A nested loop does the same work: finish one row, then move to the next.

A double sum adds one row at a time

Hold i fixed while the inner index j runs through its values.

$$\sum_{i=1}^{4}\sum_{j=1}^{3} ij$$

Inner sum

$$i + 2i + 3i = 6i$$

Outer sum

$$6 + 12 + 18 + 24 = 60$$

	j = 1	j = 2	j = 3	Inner sums
i = 1	1	2	3	→ 6
i = 2	2	4	6	
i = 3	3	6	9	
i = 4	4	8	12	

$$i = 1: \quad 1 + 2 + 3 = 6$$

A nested loop does the same work: finish one row, then move to the next.

A double sum adds one row at a time

Hold i fixed while the inner index j runs through its values.

$$\sum_{i=1}^{4}\sum_{j=1}^{3} ij$$

Inner sum

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Outer sum

$$6 + 12 + 18 + 24 = 60$$

	j = 1	j = 2	j = 3		Inner sums
i = 1	1	2	3	→	6
i = 2	2	4	6	→	12
i = 3	3	6	9	→	18
i = 4	4	8	12	→	24

$$i = 4: \quad 4 + 8 + 12 = 24$$

A nested loop does the same work: finish one row, then move to the next.

A double sum adds one row at a time

Hold i fixed while the inner index j runs through its values.

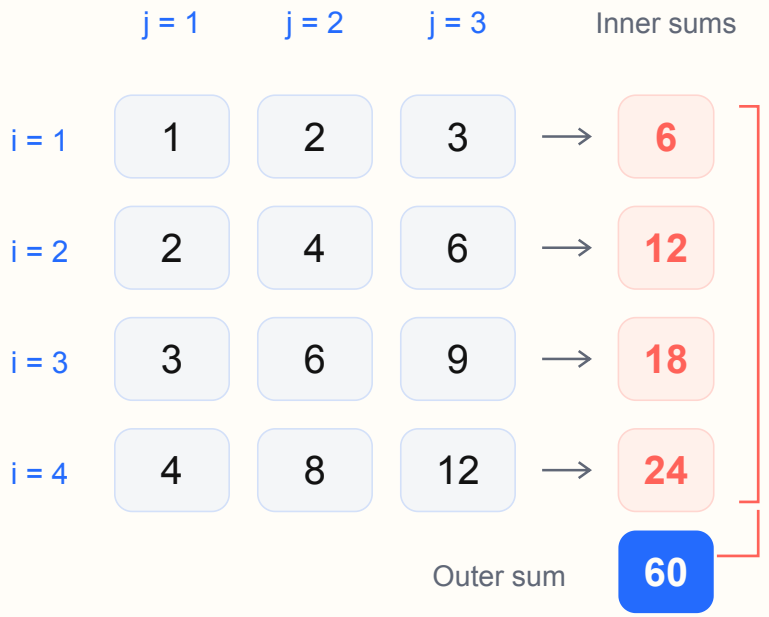
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$$6 + 12 + 18 + 24 = 60$$

A nested loop does the same work: finish one row, then move to the next.

Product notation multiplies the terms

The bounds work just as they do for a sum.

Add

$$\sum_{k=1}^{4} k = 1 + 2 + 3 + 4 = 10$$

Multiply

$$\prod_{k=1}^{4} k = 1 \cdot 2 \cdot 3 \cdot 4 = 24$$

The bounds specify the terms; the symbol specifies the operation.

Pairing the ends gives the arithmetic sum

$S = a_1 + \cdots + a_n$, with a constant difference.

Reverse the terms

$$S = a_n + \cdots + a_1$$

Add the two rows

$$2S = n(a_1 + a_n)$$

Divide by 2

$$S = \frac{n(a_1 + a_n)}{2}$$



$$S = 3 + 5 + 7 + 9 + 11 + 13 + 15$$

Number of terms × average of the endpoints.

Pairing the ends gives the arithmetic sum

$S = a_1 + \cdots + a_n$, with a constant difference.

Reverse the terms

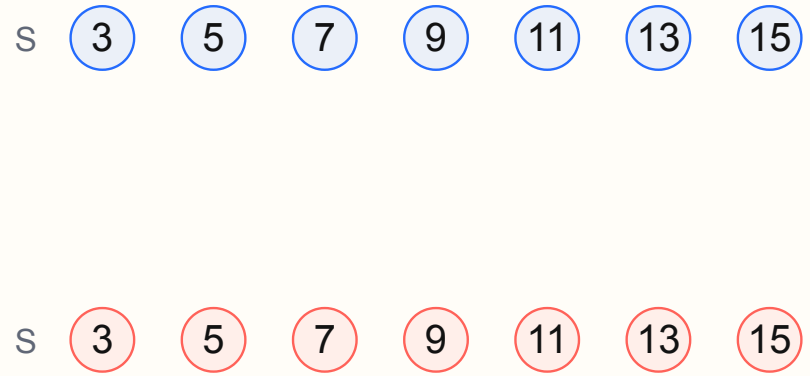
$$S = a_n + \cdots + a_1$$

Add the two rows

$$2S = n(a_1 + a_n)$$

Divide by 2

$$S = \frac{n(a_1 + a_n)}{2}$$



Make a second copy of the sum

Number of terms × average of the endpoints.

Pairing the ends gives the arithmetic sum

$S = a_1 + \cdots + a_n$, with a constant difference.

Reverse the terms

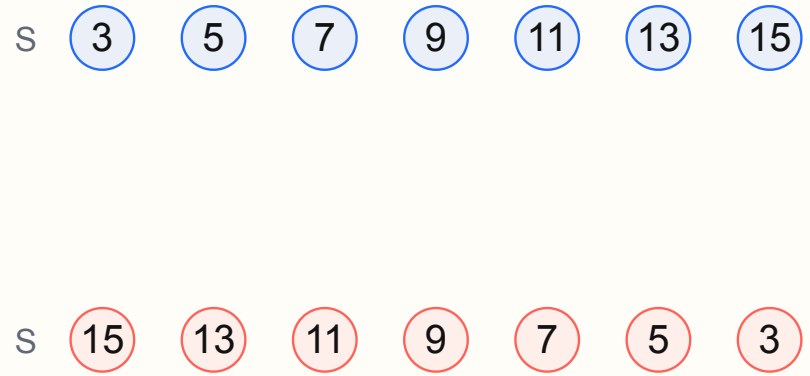
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Number of terms × average of the endpoints.

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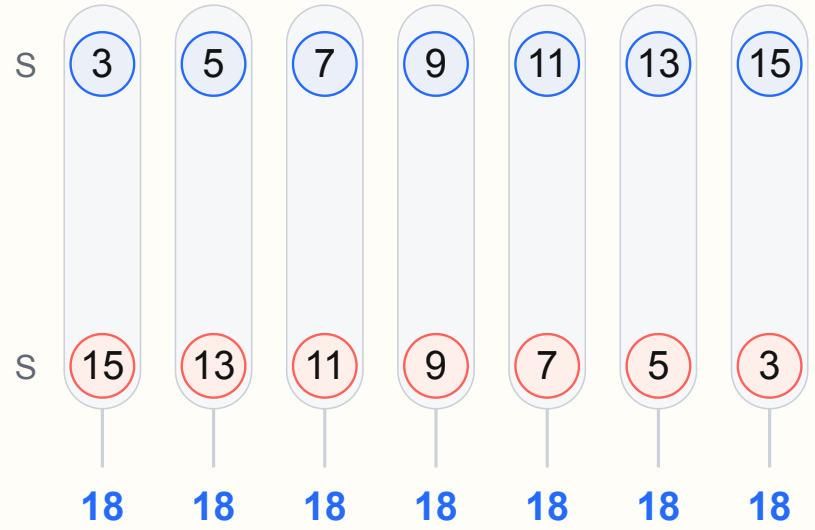
$$S = a_n + \cdots + a_1$$

Add the two rows

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Divide by 2

$$S = \frac{n(a_1 + a_n)}{2}$$



Every column has the same total: 18

Number of terms × average of the endpoints.

Pairing the ends gives the arithmetic sum

$S = a_1 + \cdots + a_n$, with a constant difference.

Reverse the terms

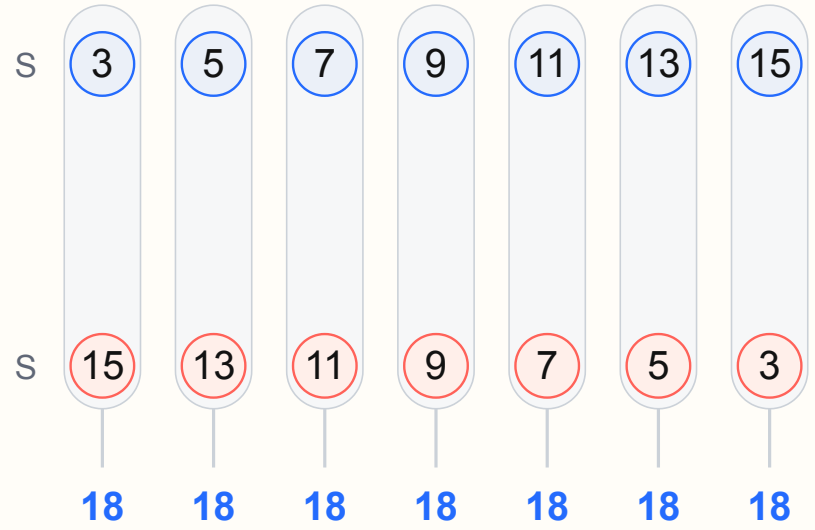
$$S = a_n + \cdots + a_1$$

Add the two rows

$$2S = n(a_1 + a_n)$$

Divide by 2

$$S = \frac{n(a_1 + a_n)}{2}$$



$$2S = 7 \cdot 18 = 126 \quad \Rightarrow \quad S = 63$$

Number of terms × average of the endpoints.

EXERCISE 8

Sum the first n positive integers

Apply the arithmetic-sum formula to $1 + 2 + \cdots + n$.

01 **First term, last term, and term count?**

1, n, **and** n.

02 **Find the sum.**

$\sum_{k=1}^{n} k = \frac{n(n+1)}{2}$.

03 **What if the indices run from 0 through n?**

$n+1$ **terms:** $\sum_{k=0}^{n} k = \frac{(n+1)(0+n)}{2} = \frac{n(n+1)}{2}$.

Two triangles make a rectangle

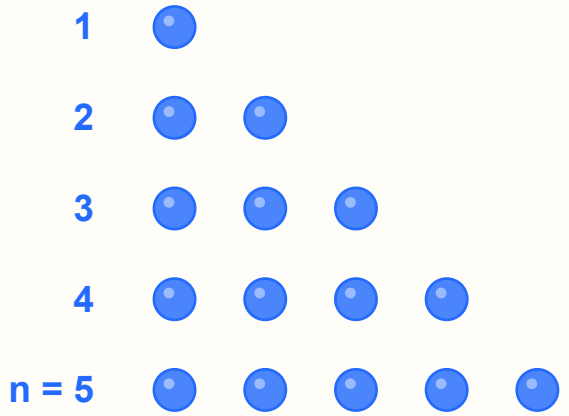
$$T_n = 1 + 2 + \cdots + n.$$

Count the rectangle

$$2T_n = n(n+1)$$

Take one of the two triangles

$$T_n = \frac{n(n+1)}{2}$$



The diagram shows $n = 5$: two copies of 15 dots make 30.

Two triangles make a rectangle

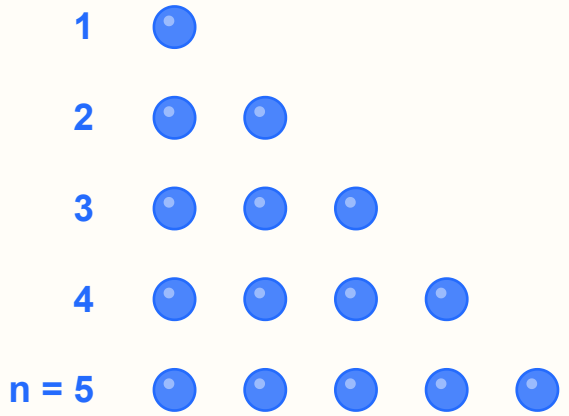
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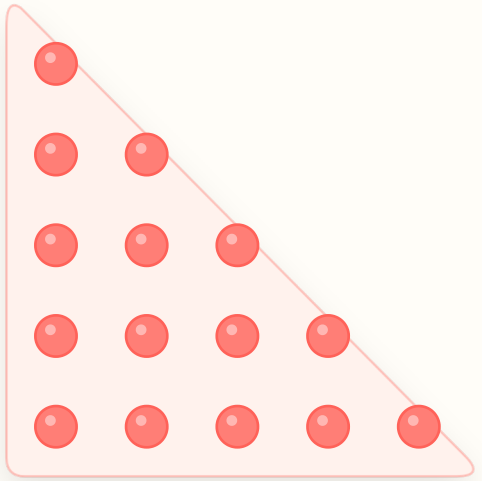
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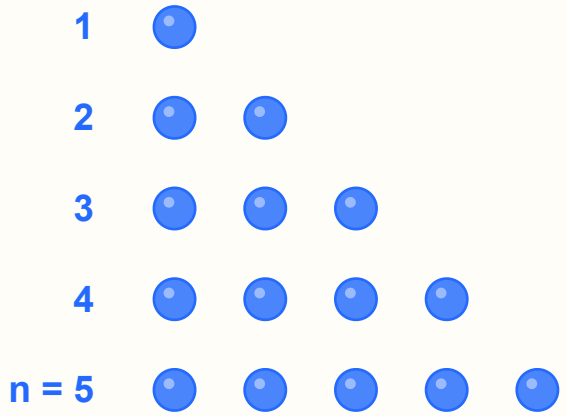
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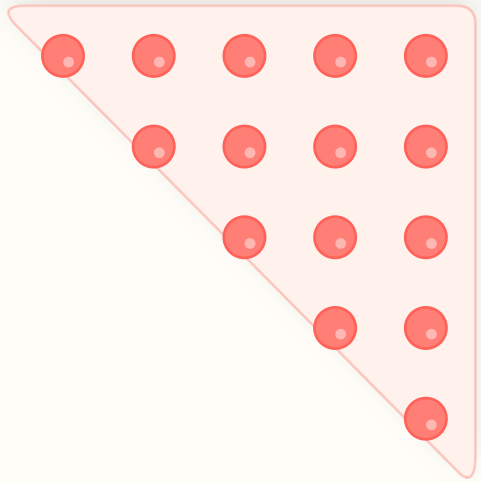
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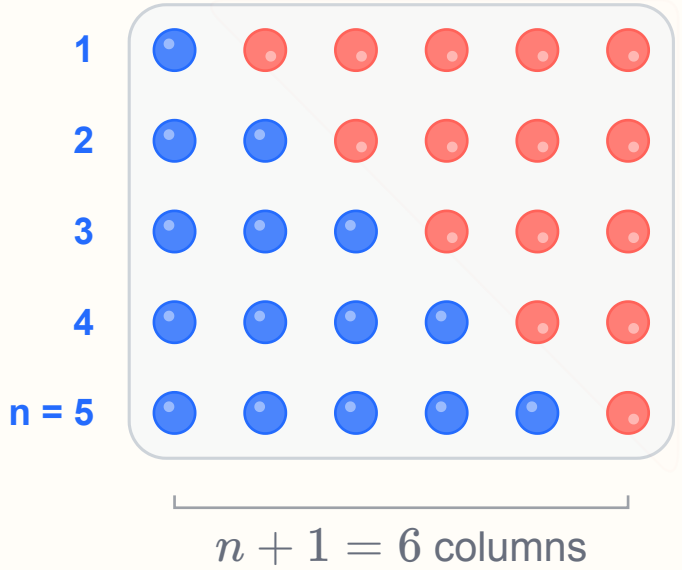
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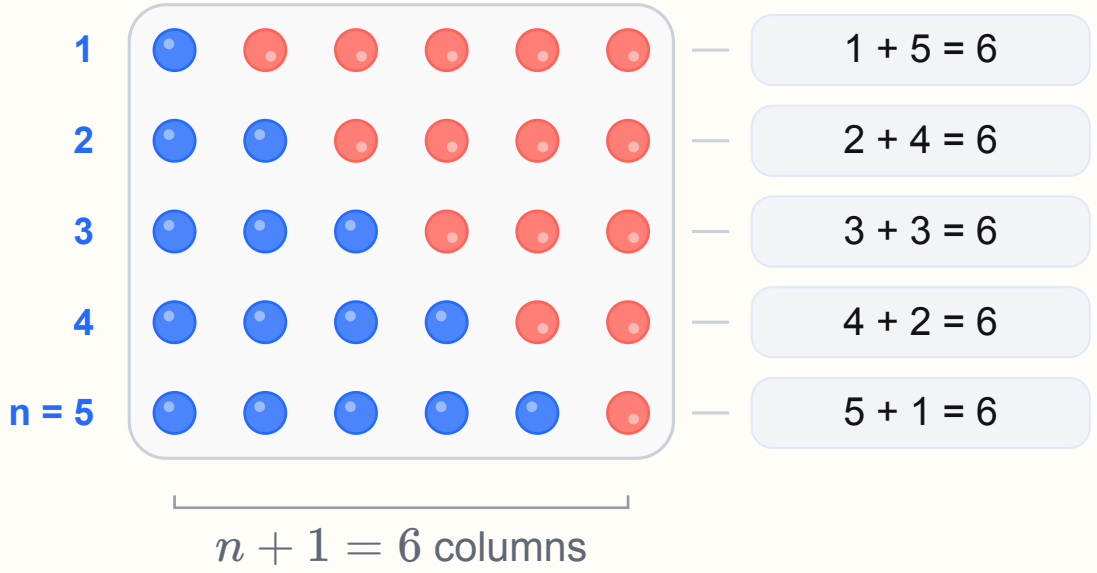
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Take one of the two triangles

$$T_n = \frac{n(n+1)}{2}$$



Two triangles $2T_n = n(n+1) = 30$

One triangle $T_n = \frac{n(n+1)}{2} = 15$

The diagram shows $n = 5$: two copies of 15 dots make 30.

Cricket: the total and the mean

25 innings: $0, 5, 10, \ldots, 120$ runs.

Total runs

$$S = \frac{25(0 + 120)}{2} = 1500$$

Mean score

$$\overline{a} = \frac{1500}{25} = 60$$

This is also the batting average if the batsman is dismissed in every innings.

Odd numbers build a square

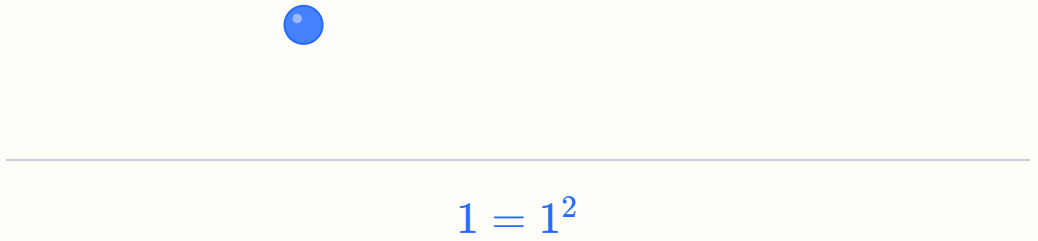
The kth border adds $2k - 1$ new dots.

A square grows by an L-shaped border

$$k^2 - (k-1)^2 = 2k - 1$$

Add the borders

$$\sum_{k=1}^{n} (2k - 1) = n^2$$



$1 = 1^2$

At $n = 7$: $1 + 3 + 5 + 7 + 9 + 11 + 13 = 49.$

Odd numbers build a square

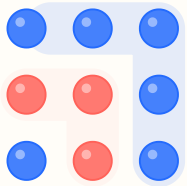
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A square grows by an L-shaped border

$$k^2 - (k-1)^2 = 2k - 1$$

Add the borders

$$\sum_{k=1}^{n} (2k - 1) = n^2$$



$$1 + 3 + 5 = 3^2$$

At $n = 7$: $1 + 3 + 5 + 7 + 9 + 11 + 13 = 49.$

Odd numbers build a square

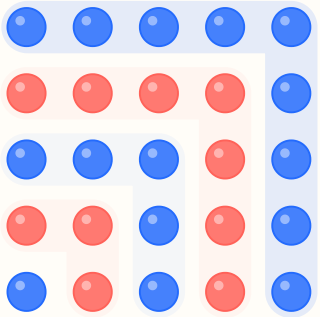
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A square grows by an L-shaped border

$$k^2-(k-1)^2=2k-1$$

Add the borders

$$\sum_{k=1}^{n}(2k-1)=n^2$$



$$1+3+5+7+9=5^2$$

At $n=7$: $1+3+5+7+9+11+13=49$.

Odd numbers build a square

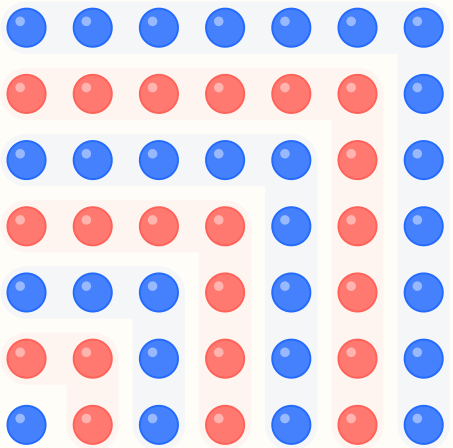
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A square grows by an L-shaped border

$$k^2-(k-1)^2=2k-1$$

Add the borders

$$\sum_{k=1}^{n}(2k-1)=n^2$$



$$1+3+\cdots+(2n-1)=n^2$$

At $n=7$: $1+3+5+7+9+11+13=49$.

Telescoping leaves only the endpoints

Expand neighboring differences and match each positive term with its negative copy.

Write the cancellation

$$\sum_{k=1}^{n}(b_k - b_{k-1})$$
$$= (b_1 - b_0) + (b_2 - b_1) + \cdots + (b_n - b_{n-1})$$

Keep the unmatched terms

$$b_n - b_0$$

For $b_k = k^2$, this also proves the sum of the first n odd numbers.

EXERCISE 9

Make the cancellation visible

Use $\dfrac{1}{k(k+1)} = \dfrac{1}{k} - \dfrac{1}{k+1}$.

01 **Expand** $\displaystyle\sum_{k=1}^{4} \frac{1}{k(k+1)}$.

$(1 - \frac{1}{2}) + (\frac{1}{2} - \frac{1}{3}) + (\frac{1}{3} - \frac{1}{4}) + (\frac{1}{4} - \frac{1}{5}) = \frac{4}{5}$.

---

02 **Find** $\displaystyle\sum_{k=1}^{n} \frac{1}{k(k+1)}$.

$1 - \dfrac{1}{n+1} = \dfrac{n}{n+1}$.

Six square pyramids fill one box

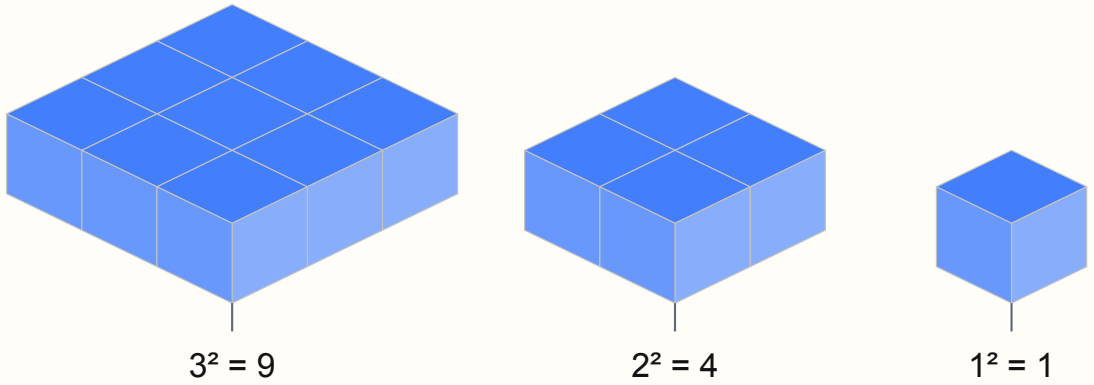
Stack square layers to make $Q_n = 1^2 + 2^2 + \cdots + n^2$.

Six congruent copies

$$6Q_n = n(n+1)(2n+1)$$

Divide the volume by six

$$Q_n = \frac{n(n+1)(2n+1)}{6}$$



$$Q_3 = 1^2 + 2^2 + 3^2 = 14$$

The picture uses $n = 3$; the same construction works for every positive integer n.

Six square pyramids fill one box

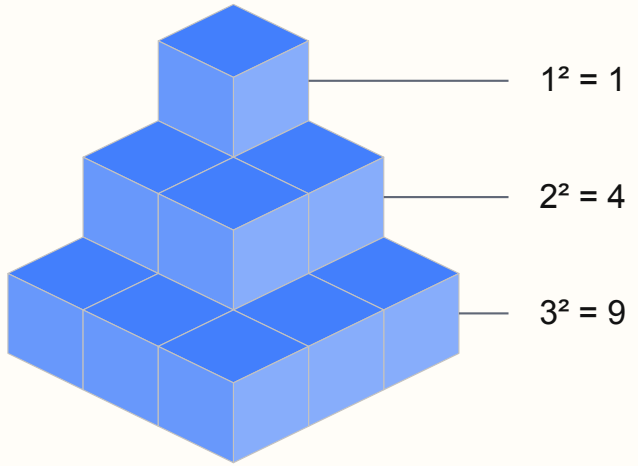
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Six square pyramids fill one box

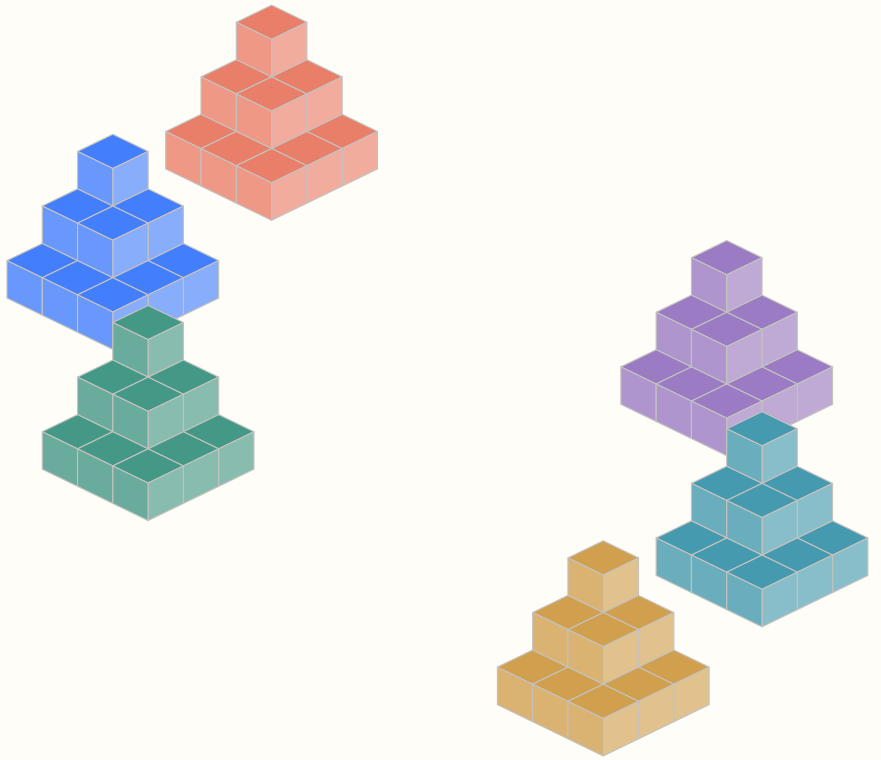
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Six congruent copies

$$6Q_n = n(n+1)(2n+1)$$

Divide the volume by six

$$Q_n = \frac{n(n+1)(2n+1)}{6}$$



$$6Q_3 = 6 \cdot 14$$

The picture uses $n = 3$; the same construction works for every positive integer n.

Six square pyramids fill one box

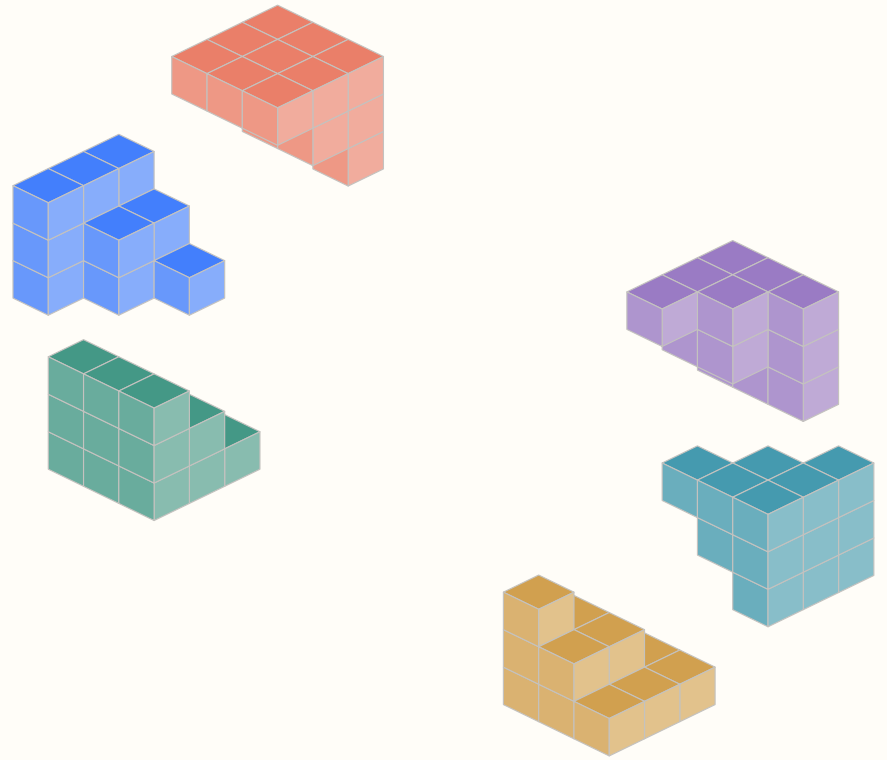
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Six congruent copies

$$6Q_n = n(n+1)(2n+1)$$

Divide the volume by six

$$Q_n = \frac{n(n+1)(2n+1)}{6}$$



$$6Q_3 = 6 \cdot 14$$

The picture uses $n = 3$; the same construction works for every positive integer n.

Six square pyramids fill one box

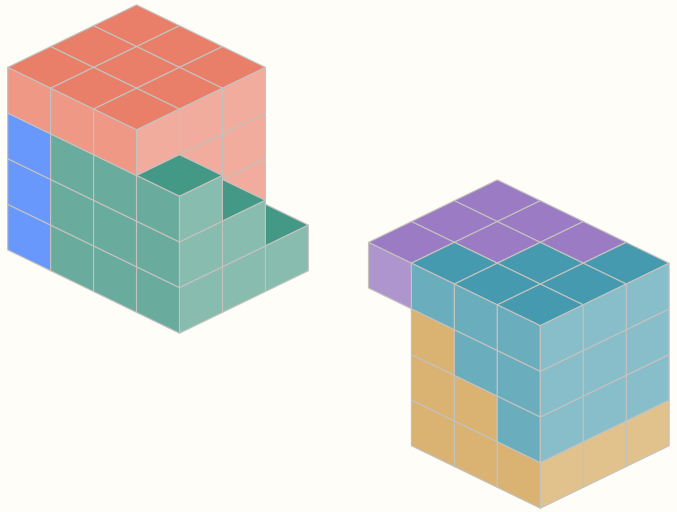
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Six congruent copies

$$6Q_n = n(n+1)(2n+1)$$

Divide the volume by six

$$Q_n = \frac{n(n+1)(2n+1)}{6}$$



$$3Q_3 + 3Q_3 = 42 + 42$$

The picture uses $n = 3$; the same construction works for every positive integer n.

Six square pyramids fill one box

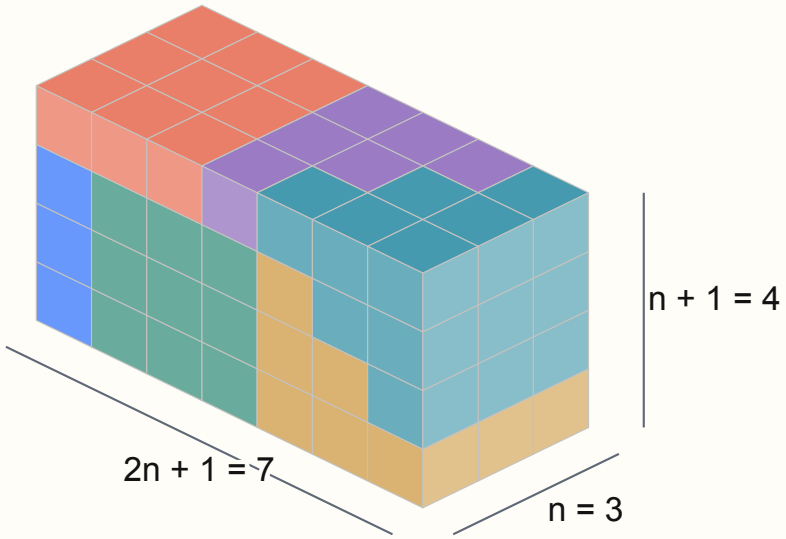
Stack square layers to make $Q_n = 1^2 + 2^2 + \cdots + n^2$.

Six congruent copies

$$6Q_n = n(n+1)(2n+1)$$

Divide the volume by six

$$Q_n = \frac{n(n+1)(2n+1)}{6}$$



$$6Q_3 = 7 \cdot 3 \cdot 4 = 84$$

The picture uses $n = 3$; the same construction works for every positive integer n.

Six square pyramids fill one box

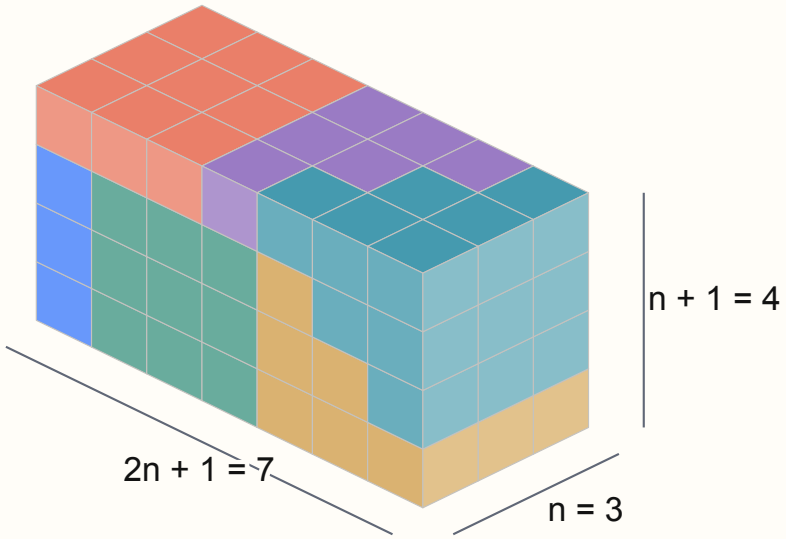
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Divide the volume by six

$$Q_n = \frac{n(n+1)(2n+1)}{6}$$



$$Q_n = \frac{n(n+1)(2n+1)}{6}$$

The picture uses $n = 3$; the same construction works for every positive integer n.

The sum of cubes is a squared triangular number

Let $T_k = k(k+1)/2$, with $T_0 = 0$.

Factor the difference

$$T_k^2 - T_{k-1}^2 = (T_k - T_{k-1})(T_k + T_{k-1})$$

Use $T_k - T_{k-1} = k$ and $T_k + T_{k-1} = k^2$

$$T_k^2 - T_{k-1}^2 = k^3$$

Sum the differences

$$\sum_{k=1}^{n} k^3 = T_n^2 = \left(\frac{n(n+1)}{2}\right)^2$$

EXERCISE 10

Remove the prefix you do not want

Use $Q_n = n(n+1)(2n+1)/6$.

01 **Express $\sum_{k=4}^{6} k^2$ using two prefix sums.**

$Q_6 - Q_3$.

02 **Evaluate it.**

$91 - 14 = 77 = 16 + 25 + 36$.

Sum of geometric sequence

$S = a + ar + \cdots + ar^{n-1}$ contains n terms.

Multiply by r

$$rS = ar + ar^2 + \cdots + ar^n$$

Subtract the matching terms

$$(1 - r)S = a - ar^n$$

Solve for $r \neq 1$

$$S = a\frac{1 - r^n}{1 - r}$$

For $r = 1$, the sum is na.

EXERCISE 11

Geometric sums: count before substituting

Identify the first term, ratio, and term count.

01 $2 + 6 + 18 + 54 + 162$

Five terms: $2(1 - 3^5)/(1 - 3) = 242$.

---

02 $1 + 1/2 + 1/4 + 1/8$

Four terms: $(1 - (1/2)^4)/(1 - 1/2) = 15/8$.

---

03 $7 + 7 + 7 + 7$

Here $r = 1$, so the sum is $4 \cdot 7 = 28$.

An infinite sum is a limit of partial sums

For $|r| < 1$, the remaining power r^n tends to zero.

Start with n terms

$$S_n = a\frac{1 - r^n}{1 - r}$$

Take the limit

$$\sum_{k=0}^{\infty} ar^k = \frac{a}{1 - r}$$

Example: keep halving

$$1 + \frac{1}{2} + \frac{1}{4} + \cdots = 2$$

Check $|r| < 1$ before using the infinite-sum formula.

Smaller terms still increase the total

The harmonic number adds terms starting at 1.

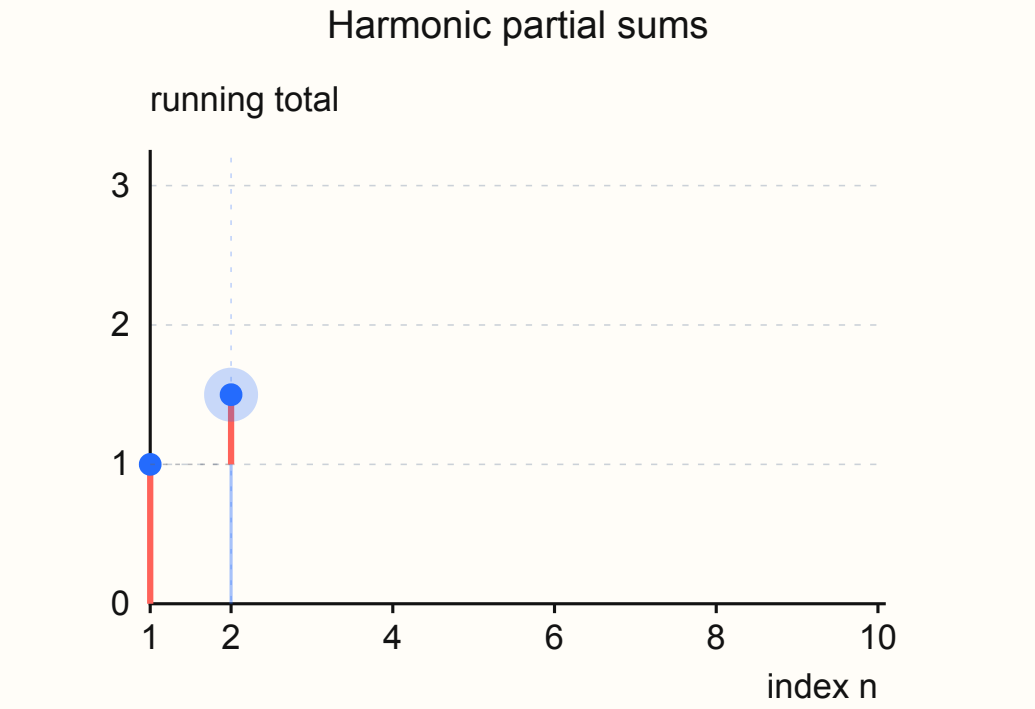
$$H_n = \sum_{k=1}^{n} \frac{1}{k}$$

First totals

$$H_1 = 1,\ H_2 = \frac{3}{2},\ H_3 = \frac{11}{6}$$

Starting at $1/2$ instead

$$\sum_{k=2}^{n} \frac{1}{k} = H_n - 1$$



$$H_2 = H_1 + \tfrac{1}{2} \approx 1.500$$

A small added term still increases the running total.

Smaller terms still increase the total

The harmonic number adds terms starting at 1.

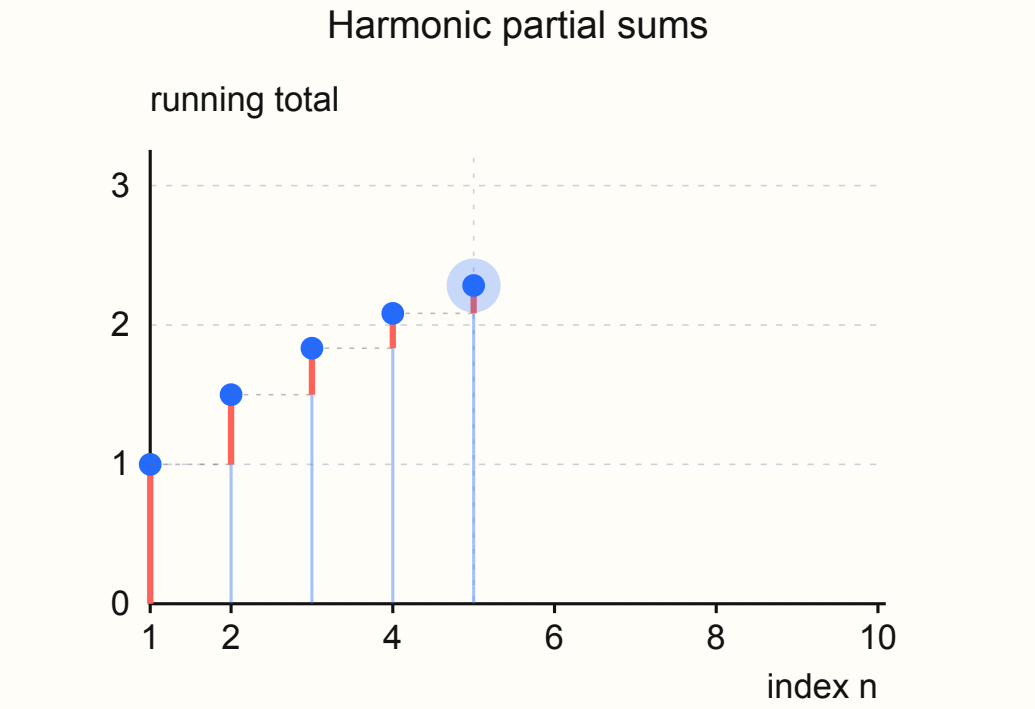
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First totals

$$H_1 = 1,\ H_2 = \frac{3}{2},\ H_3 = \frac{11}{6}$$

Starting at $1/2$ instead

$$\sum_{k=2}^{n} \frac{1}{k} = H_n - 1$$



$$H_5 = H_4 + \tfrac{1}{5} \approx 2.283$$

A small added term still increases the running total.

Smaller terms still increase the total

The harmonic number adds terms starting at 1.

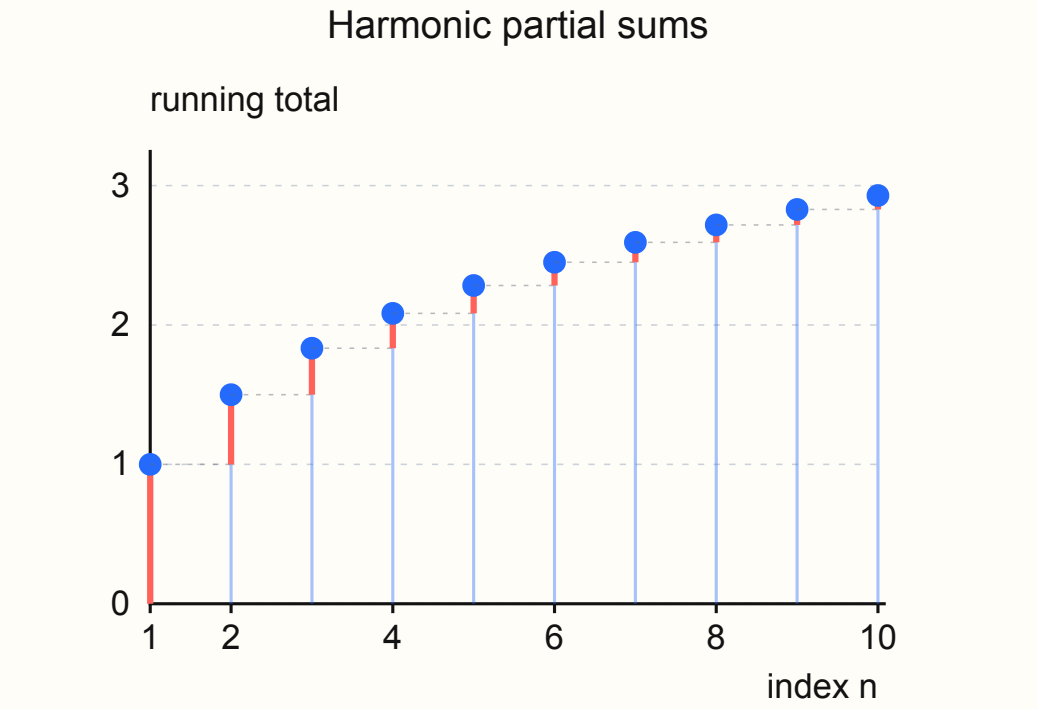
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First totals

$$H_1 = 1,\ H_2 = \frac{3}{2},\ H_3 = \frac{11}{6}$$

Starting at $1/2$ instead

$$\sum_{k=2}^{n} \frac{1}{k} = H_n - 1$$



$H_n - H_{n-1} = 1/n > 0$: the total keeps growing.

A small added term still increases the running total.

Terms tending to zero are not enough

Each doubling block of the harmonic sum contributes at least $1/2$.

Group the terms

$$H_8 = 1 + \frac{1}{2} + (\frac{1}{3} + \frac{1}{4}) + (\frac{1}{5} + \cdots + \frac{1}{8})$$

Bound each block from below

$$H_{2^m} \geq 1 + \frac{m}{2}$$

The geometric sum converges; the harmonic sum grows without bound.

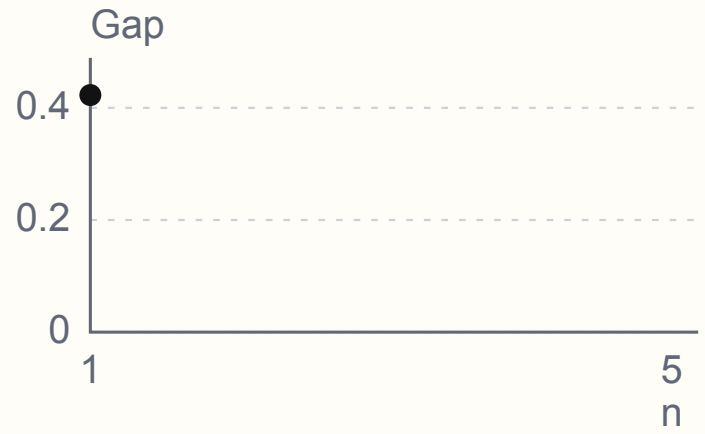
The logarithmic approximation gets closer

Exact: $H_n = \sum_{k=1}^{n} 1/k$ Approximation: $\ln n + \gamma$ $\gamma \approx 0.57721$



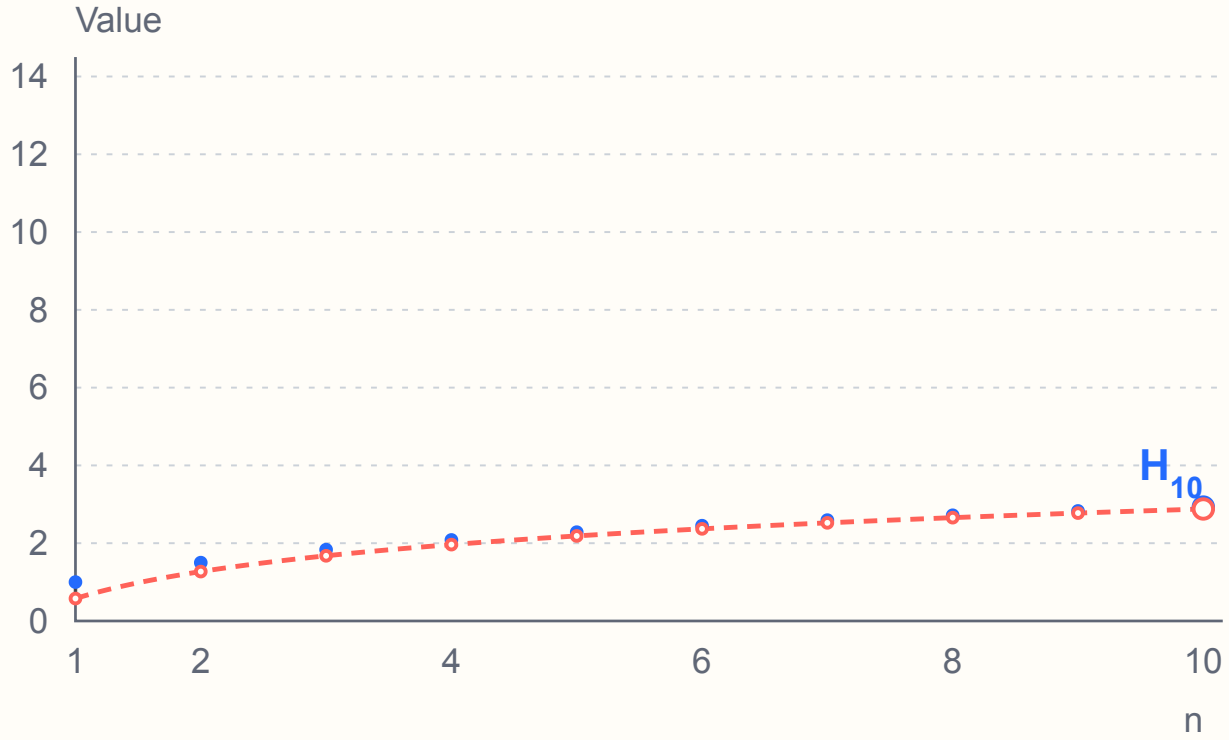
Terms revealed: **1**

H_n	**1.0000000**
$\ln n + \gamma$	**0.5772157**
Gap	**0.4227843**



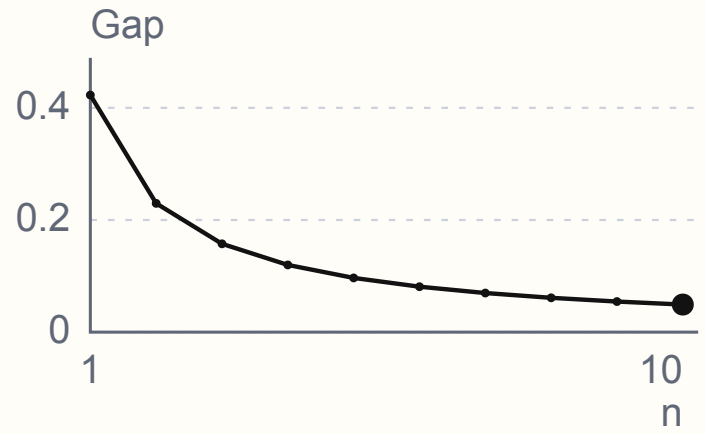
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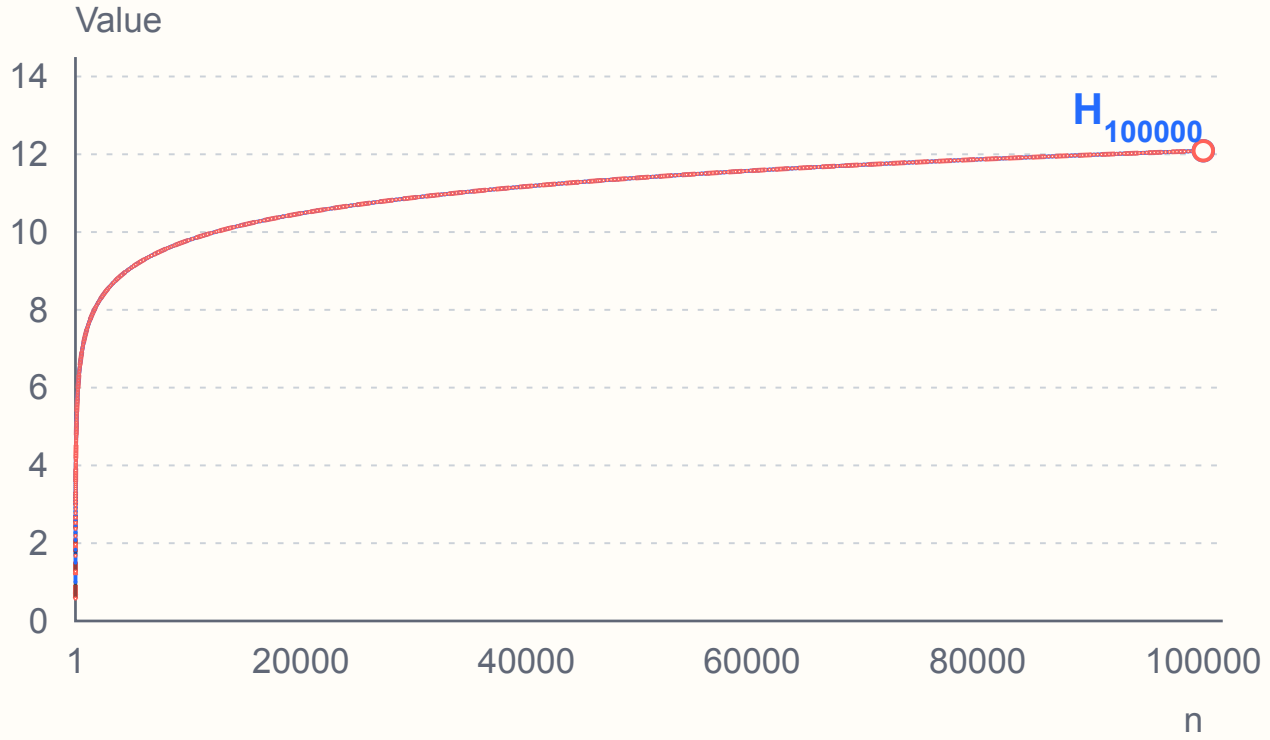
Terms revealed: **10**

H_n	**2.9289683**
$\ln n + \gamma$	**2.8798008**
Gap	**0.0491675**



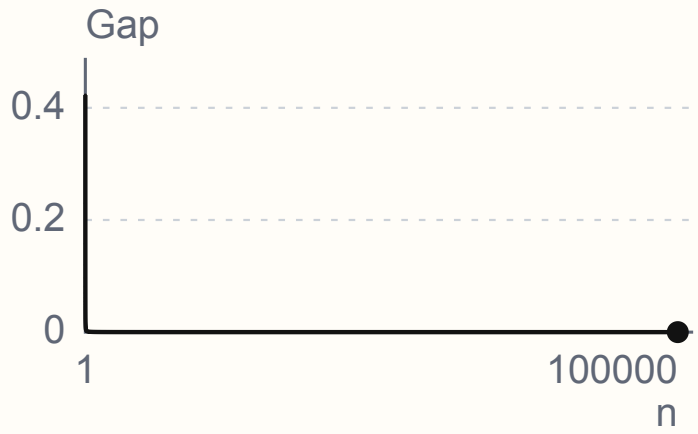
The logarithmic approximation gets closer

Exact: $H_n = \sum_{k=1}^{n} 1/k$ Approximation: $\ln n + \gamma$ $\gamma \approx 0.57721$



Terms revealed: **100000**

H_n	**12.0901461**
$\ln n + \gamma$	**12.0901411**
Gap	**0.0000050**



Choose a method before a formula

Read the bounds, identify the pattern, then choose a useful transformation.

Structure	Method or result
Arithmetic: n terms	$n(a_1 + a_n)/2$
Geometric: n terms, $r \neq 1$	$a_1(1 - r^n)/(1 - r)$
Differences: $b_k - b_{k-1}$	Telescope to $b_n - b_0$
Shifted or combined terms	Reindex; use linearity; subtract a prefix.
Infinite geometric, $\lvert r \rvert < 1$	$a/(1 - r)$

For a geometric sum with $r = 1$, add n copies of the first term.

EXERCISE 12

Exit check: a term and a total

Geometric sequence: $3, 6, 12, 24, \ldots$, starting at a_1.

01 **Find the ratio and a formula for a_n.**

$r = 2$ **and** $a_n = 3 \cdot 2^{n-1}$.

---

02 **Find a_{10}.**

$3 \cdot 2^9 = 1536.$

---

03 **Find the sum of the first ten terms.**

$3(2^{10} - 1) = 3069.$

TAKEAWAY

A term gives one score; a sum gives the total

Starting at $a_1 = 0$ and adding 5 runs each innings:

$$a_{25} = 120 \qquad \sum_{k=1}^{25} a_k = 1500$$